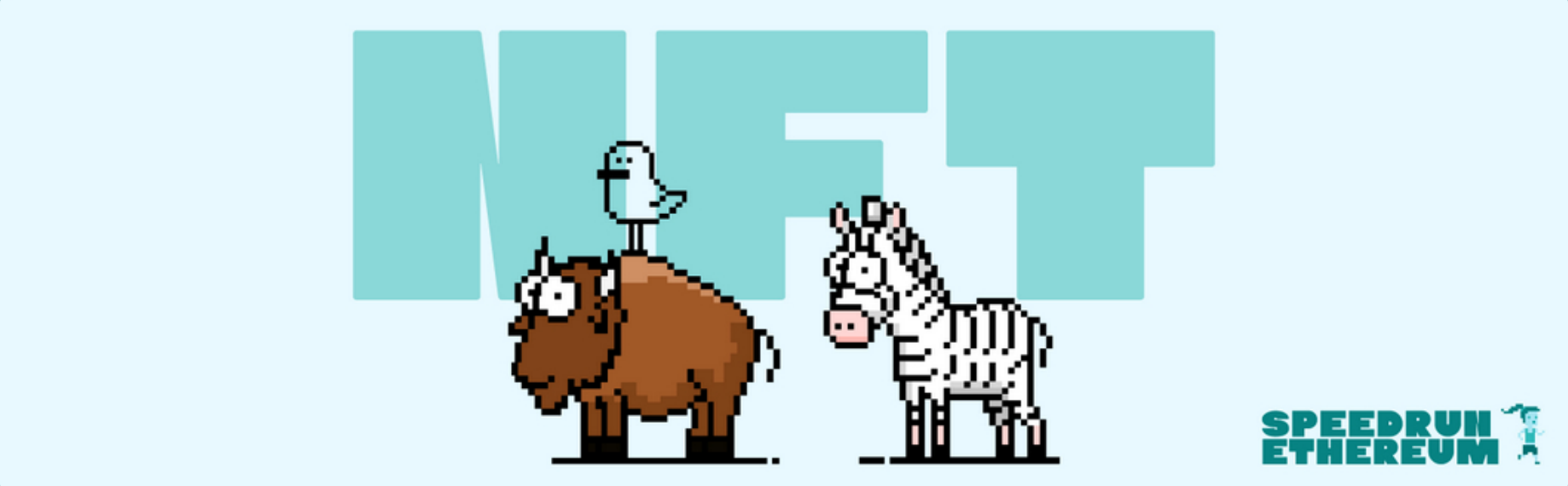
🚩 Challenge #0: 🎫 Simple NFT Example


NFT
SPEEDRUN ETHEREUM

📚 This tutorial is meant for developers that already understand the 🖍️ basics.

🧑‍🏫 If you would like a more gentle introduction for developers, watch our 15 video 🎥 Web2 to Web3 series.

🎫 Create a simple NFT:

👷 You'll compile and deploy your first smart contracts. Then, you'll use a template NextJS app full of important Ethereum components and hooks. Finally, you'll deploy an NFT to a public network to share with friends! 🚀

🌟 The final deliverable is an app that lets users purchase and transfer NFTs. Deploy your contracts to a testnet, then build and upload your app to a public web server. Submit the url on SpeedRunEthereum.com!

💬 Meet other builders working on this challenge and get help in the Challenge 0 Telegram!

🤖 If you have any question during your Challenge, you can try out the Challenge AI assistant, and get answers to your Challenge/Scaffold-ETH questions. Please reach us in Telegram if something feels wrong!

Checkpoint 0: 📦 Environment 📚

Before you begin, you need to install the following tools:

- Node (>= v18.18)
- Yarn (v1 or v2+)
- Git

Then download the challenge to your computer and install dependencies by running:

```
npx create-eth@0.1.0 -e challenge-0-simple-nft challenge-0-simple-nft
cd challenge-0-simple-nft
```

> *"in the same terminal, start your local network (a local instance of a blockchain):"*

```
yarn chain
```

> *"in a second terminal window, 🛰 deploy your contract (locally):"*

```
cd challenge-0-simple-nft
yarn deploy
```

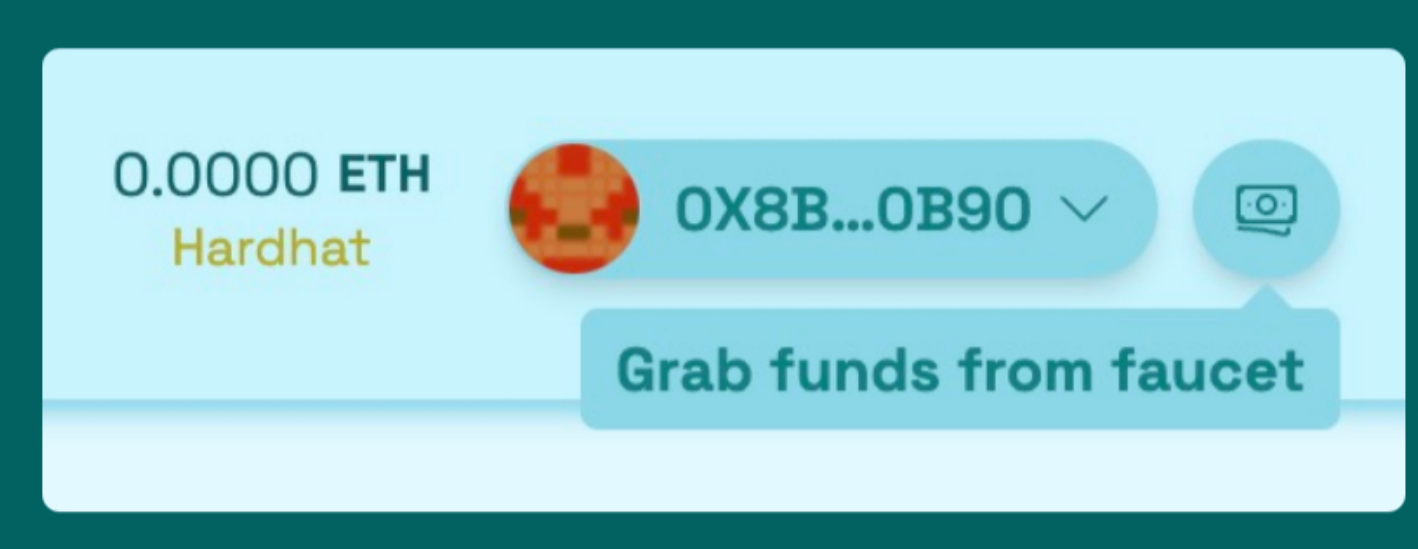
> *"in a third terminal window, start your 📱 frontend:"*

```
cd challenge-0-simple-nft
yarn start
```

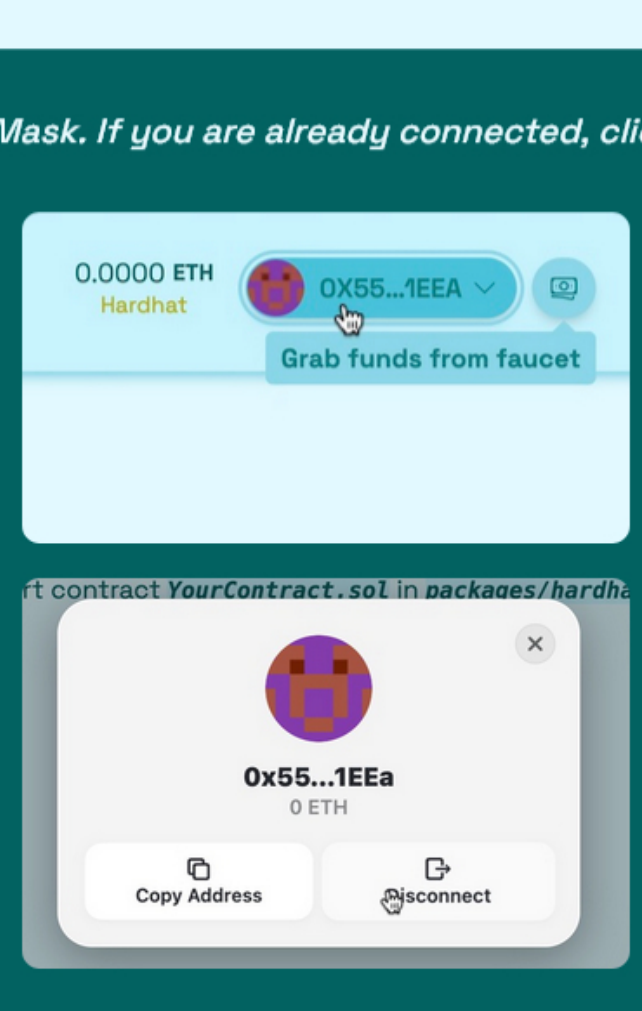
📱 Open http://localhost:3000 to see the app.

Checkpoint 1: ⛽️ Gas & Wallets 🧮

> *"⛽️ You'll need to get some funds from the faucet for gas."*

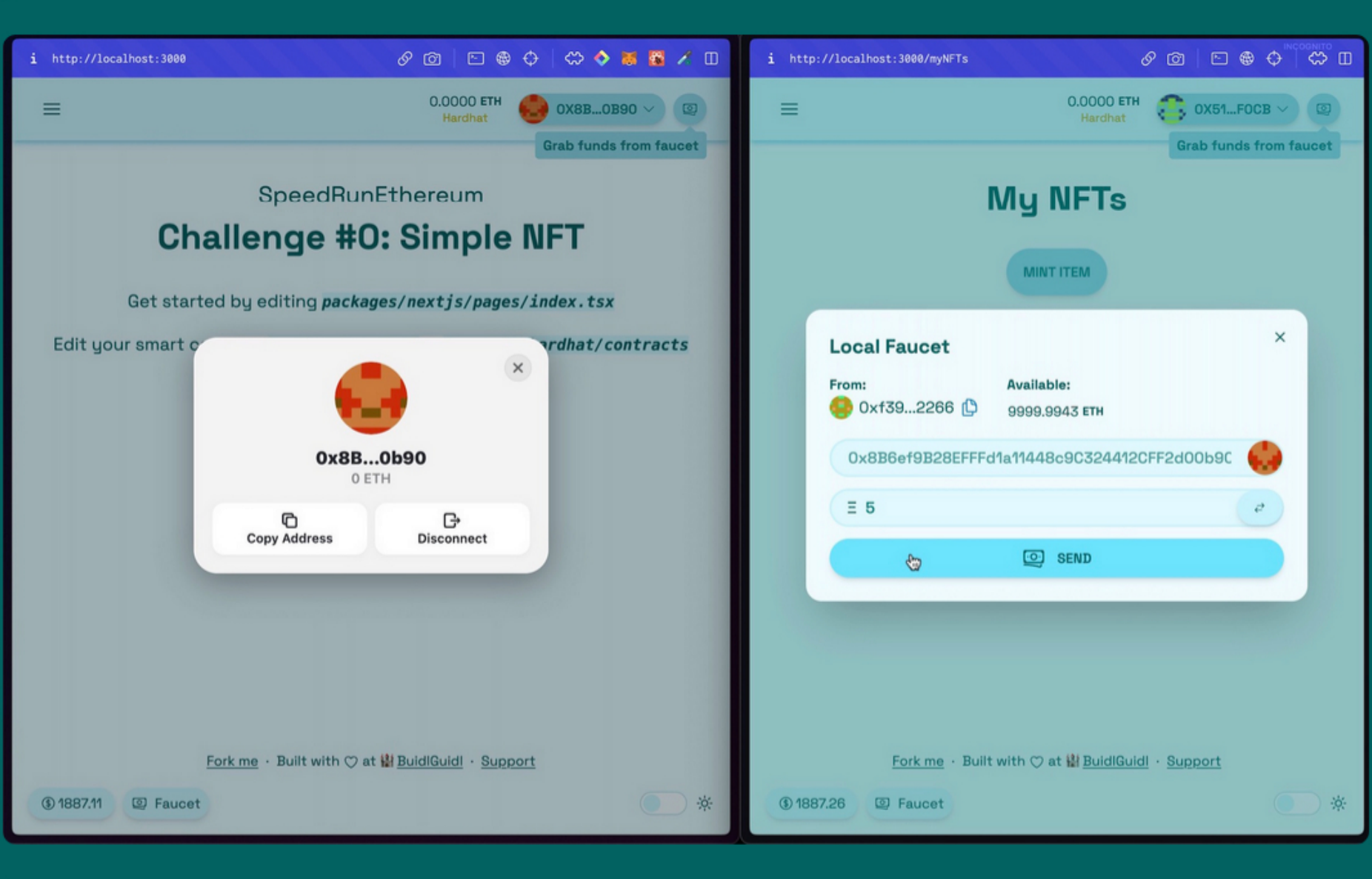

0.0000 ETH
Hardhat
0X8B...0B90
Grab funds from faucet

> *"🦊 At first, **don't** connect MetaMask. If you are already connected, click **Disconnect**:"*


0.0000 ETH
Hardhat
0X55...1EEA
Grab funds from faucet
0x55...1EEa
0 ETH
Copy Address
Disconnect

> *"🔥 We'll use burner wallets on localhost."*

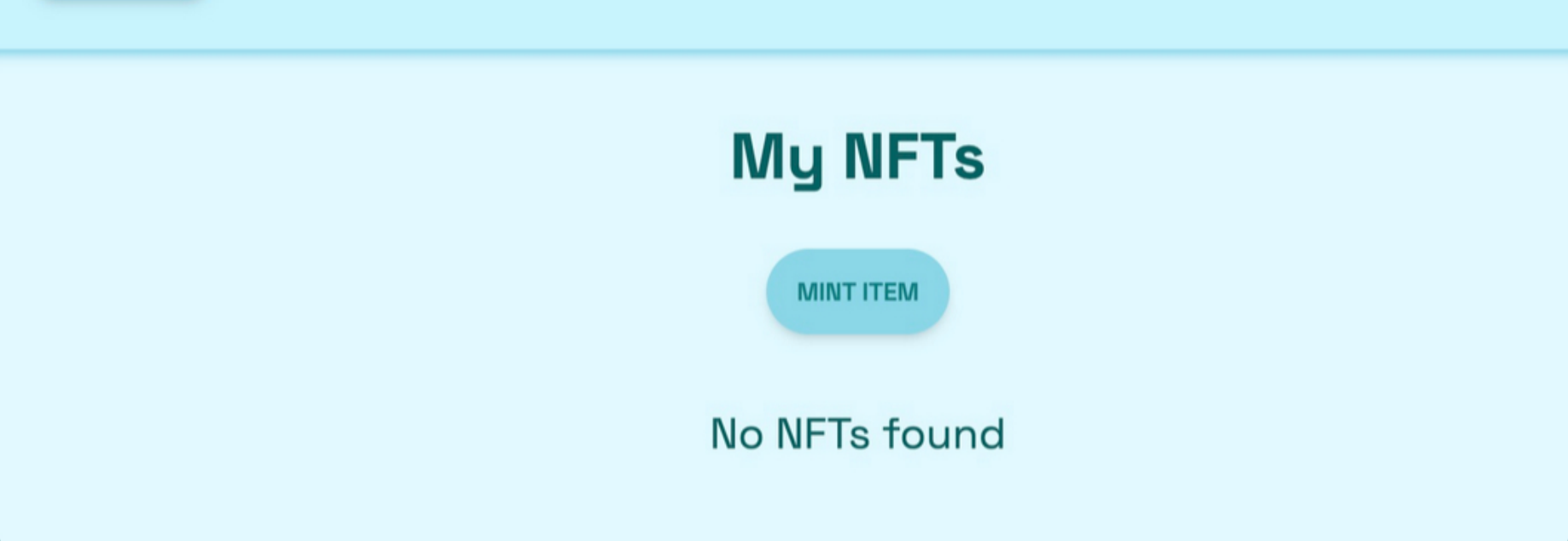
> *"👛 Explore how burner wallets work in 🏗 Scaffold-ETH 2 by opening a new incognito window and navigate to http://localhost:3000. You'll notice it has a new wallet address in the top right. Copy the incognito browser's address and send localhost test funds to it from your first browser (using the **Faucet** button in the bottom left):"*


http://localhost:3000
0.0000 ETH
Hardhat
0X8B...0B90
Grab funds from faucet
SpeedRunEthereum
Challenge #0: Simple NFT
Get started by editing packages/nextjs/pages/index.tsx
0x8B...0b90
0 ETH
Copy Address
Disconnect
Fork me · Built with ♡ at BuidlGuidl · Support
1887.11
Faucet
http://localhost:3000/myNFTs
0X51...F0CB
My NFTs
MINT ITEM
Local Faucet
From:
0xf39...2266
Available:
9999.9943 ETH
0x8B6ef9B28EFFFd1a11448c9C324412CFF2d00b90
5
SEND
1887.26

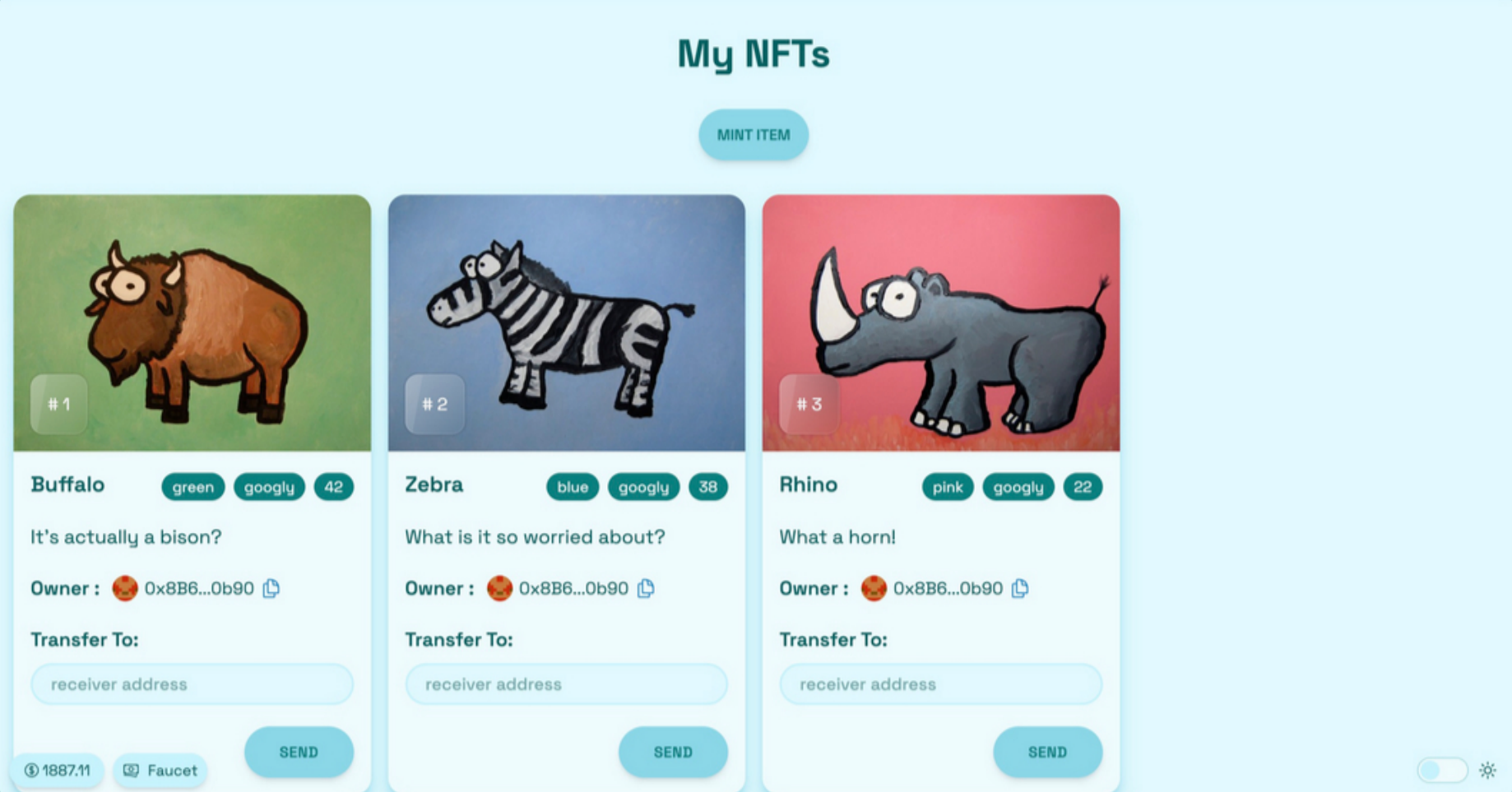
> *"🚨 When you close the incognito window, the account is gone forever. Burner wallets are great for local development but you'll move to more permanent wallets when you interact with public networks."*

Checkpoint 2: 🖨 Minting

> *"✏️ Mint some NFTs! Click the **MINT NFT** button in the `My NFTs` tab."*

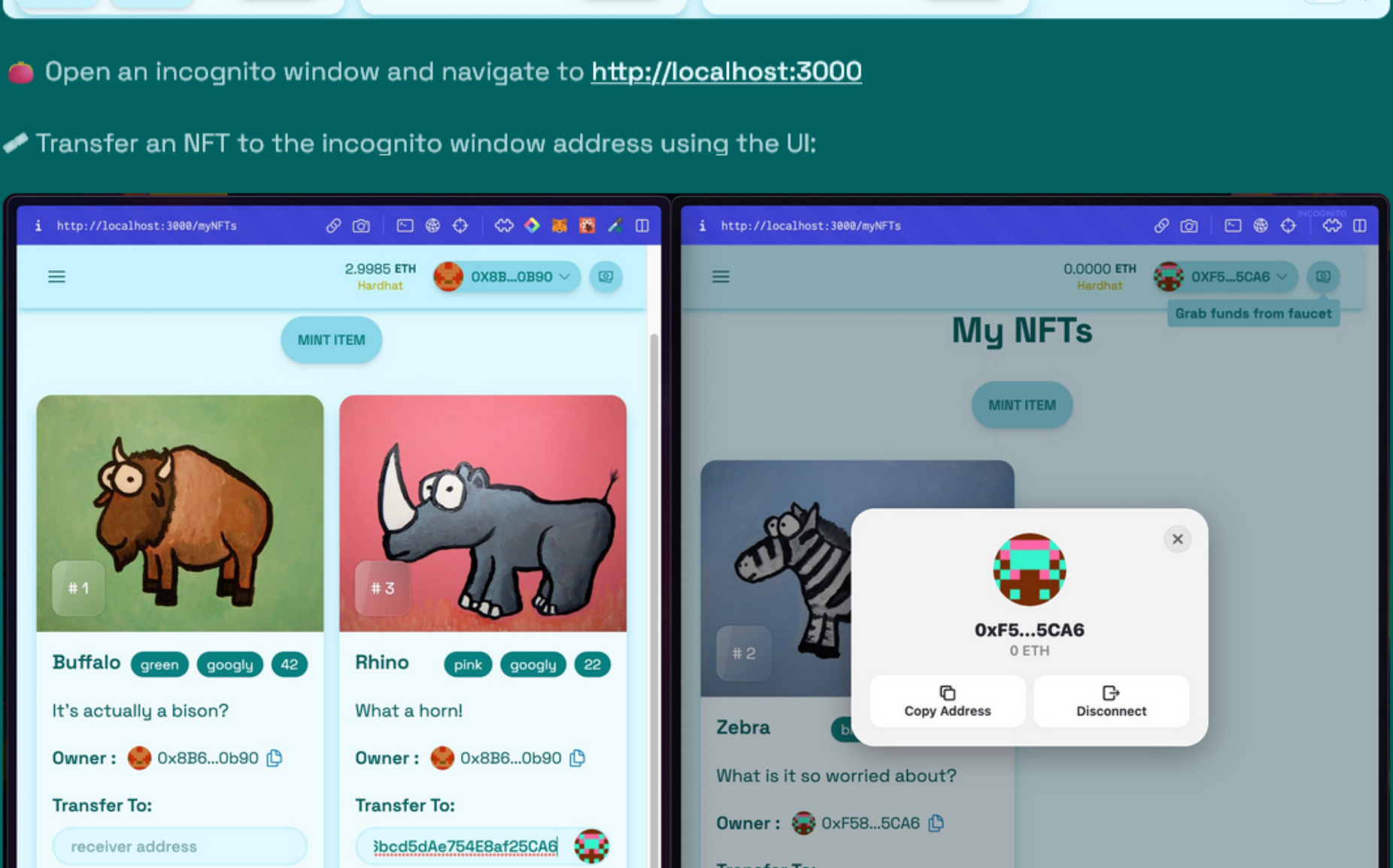

My NFTs
Transfers
IPFS Upload
IPFS Download
Debug Contracts
Block Explorer
My NFTs
MINT ITEM
No NFTs found

👀 You should see your NFTs start to show up:


My NFTs
MINT ITEM
#1
Buffalo
green
googly
42
It's actually a bison?
Owner :
0x8B6...0b90
Transfer To:
receiver address
SEND
#2
Zebra
blue
googly
38
What is it so worried about?
#3
Rhino
pink
22
What a horn!
1887.11
Faucet

👛 Open an incognito window and navigate to http://localhost:3000

✏️ Transfer an NFT to the incognito window address using the UI:


http://localhost:3000/myNFTs
2.9985 ETH
Hardhat
0X8B...0B90
MINT ITEM
Buffalo
It's actually a bison?
Owner :
0x8B6...0b90
Transfer To:
receiver address
SEND
Rhino
What a horn!
3bcd5dAe754E8af25CA6
1872.37
Faucet
0.0000 ETH
0XF5...5CA6
Grab funds from faucet
My NFTs
0xF5...5CA6
0 ETH
Copy Address
Disconnect
Zebra
What is it so worried about?
0xF58...5CA6

👛 Try to mint an NFT from the incognito window.

> *"Can you mint an NFT with no funds in this address? You might need to grab funds from the faucet to pay for the gas!"*

🕵️ Inspect the `Debug Contracts` tab to figure out what address is the owner of YourCollectible?

🔏 You can also check out your smart contract `YourCollectible.sol` in `packages/hardhat/contracts`.

💼 Take a quick look at your deploy script `00_deploy_your_contract.js` in `packages/hardhat/deploy`.

📝 If you want to edit the frontend, navigate to `packages/nextjs/app` and open the specific page you want to modify. For instance: `/myNFTs/page.tsx`. For guidance on routing and configuring pages/layouts checkout the Next.js documentation.

Checkpoint 3: 💾 Deploy your contract! 🛰

🛰 Ready to deploy to a public testnet?!?

> *"Change the defaultNetwork in `packages/hardhat/hardhat.config.ts` to `sepolia`."*

```
const config: HardhatUserConfig = {
  solidity: "0.8.2",
  defaultNetwork: "sepolia",
```

🔐 Generate a deployer address with `yarn generate`. This creates a unique deployer address and saves the mnemonic locally.

> *"This local account will deploy your contracts, allowing you to avoid entering a personal private key."*

```
carletex@xfly:~/projects/se2-challenges (challenge-0-simple-nft)$ yarn generate
👛 Generating new Wallet
📄 Private Key saved to packages/hardhat/.env file
```

👩‍🚀 Use `yarn account` to view your deployer account balances.

```
carletex@xfly:~/projects/se2-challenges (challenge-0-simple-nft)$ yarn account
```

⛽️ You will need to send ETH to your deployer address with your wallet, or get it from a public faucet of your chosen network.

> *"Some popular Sepolia faucets are the Alchemy Faucet, Infura Faucet, and Google Cloud Faucet."*

> *"⚔️ Side Quest: Keep a 🧑‍🎤 punkwallet.io on your phone's home screen and keep it loaded with testnet eth. 🧙‍♂️ You'll look like a wizard when you can fund your deployer address from your phone in seconds."*

🚀 Deploy your NFT smart contract with `yarn deploy`.

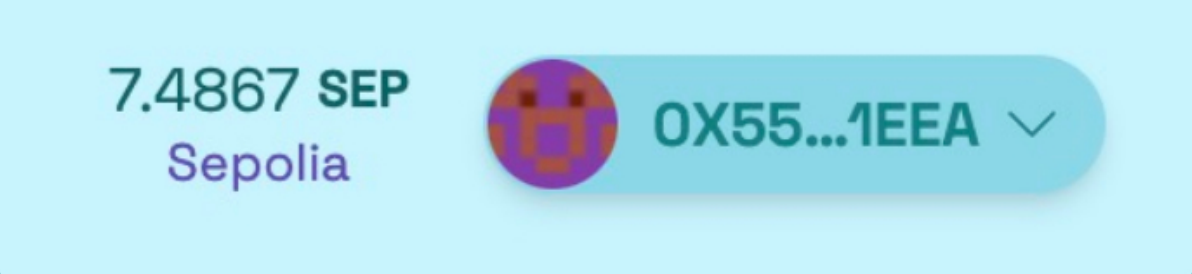
> *"💬 Hint: You can set the `defaultNetwork` in `hardhat.config.ts` to `sepolia` OR you can `yarn deploy --network sepolia`."*

Checkpoint 4: 🚢 Ship your frontend! 🚁

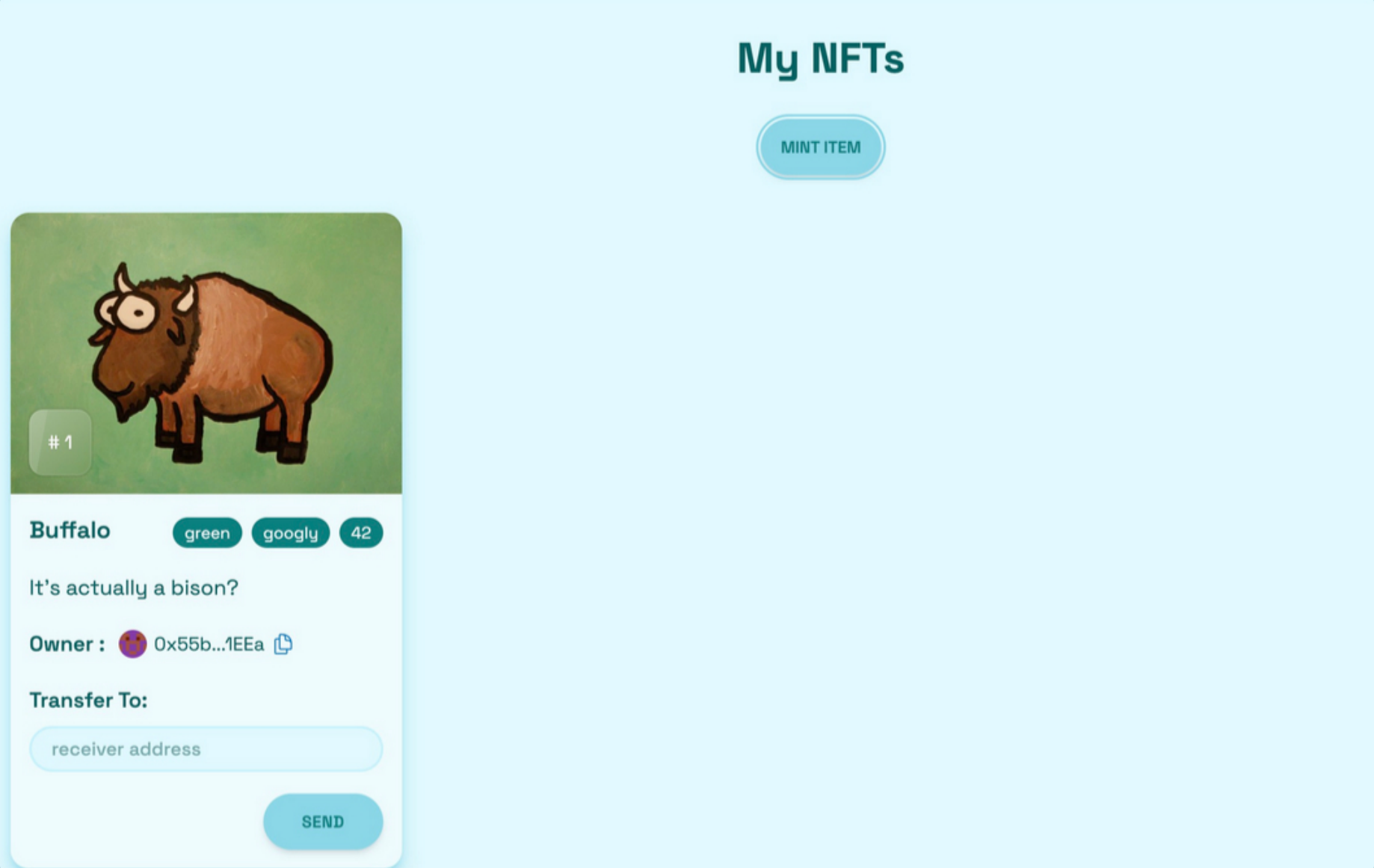
> *"✏️ Edit your frontend config in `packages/nextjs/scaffold.config.ts` to change the `targetNetwork` to `chains.sepolia`:"*

```
const scaffoldConfig = {
  // The networks on which your DApp is live
  targetNetworks: [chains.sepolia],
```

> *"You should see the correct network in the frontend (http://localhost:3000):"*


7.4867 SEP
Sepolia
0X55...1EEA

> *"🦊 Since we have deployed to a public testnet, you will now need to connect using a wallet you own or use a burner wallet. By default 🔥 `burner wallets` are only available on `hardhat`. You can enable them on every chain by setting `onlyLocalBurnerWallet: false` in your frontend config (`scaffold.config.ts` in `packages/nextjs/`)"*


My NFTs
MINT ITEM
#1
Buffalo
green
googly
42
It's actually a bison?
Owner :
0x55b...1EEa
Transfer To:
receiver address
SEND

> *"💬 Hint: For faster loading of your transfer page, consider updating the `fromBlock` passed to `useScaffoldEventHistory` in `packages/nextjs/app/transfers/page.tsx` to `blocknumber - 10` at which your contract was deployed. Example: `fromBlock: 3750241n` (where `n` represents its a BigInt). To find this blocknumber, search your contract's address on Etherscan and find the `Contract Creation` transaction line."*

🚀 Deploy your NextJS App

```
yarn vercel
```

> *"You might need to log in to Vercel first by running `yarn vercel:login`. Once you log in (email, GitHub, etc), the default options should work."*

> *"If you want to redeploy to the same production URL you can run `yarn vercel --prod`. If you omit the `--prod` flag it will deploy it to a preview/test URL."*

> *"Follow the steps to deploy to Vercel. It'll give you a public URL."*

⚠️ Run the automated testing function to make sure your app passes

```
yarn test
```

Configuration of Third-Party Services for Production-Grade Apps.

By default, 🏗 Scaffold-ETH 2 provides predefined API keys for popular services such as Alchemy and Etherscan. This allows you to begin developing and testing your applications more easily, avoiding the need to register for these services. This is great to complete your **SpeedRunEthereum**.

For production-grade applications, it's recommended to obtain your own API keys (to prevent rate limiting issues). You can configure these at:

- 🔷 `ALCHEMY_API_KEY` variable in `packages/hardhat/.env` and `packages/nextjs/.env.local`. You can create API keys from the Alchemy dashboard.
- 📃 `ETHERSCAN_API_KEY` variable in `packages/hardhat/.env` with your generated API key. You can get your key here.

> *"💬 Hint: It's recommended to store env's for nextjs in Vercel/system env config for live apps and use .env.local for local testing."*

Checkpoint 5: 📜 Contract Verification

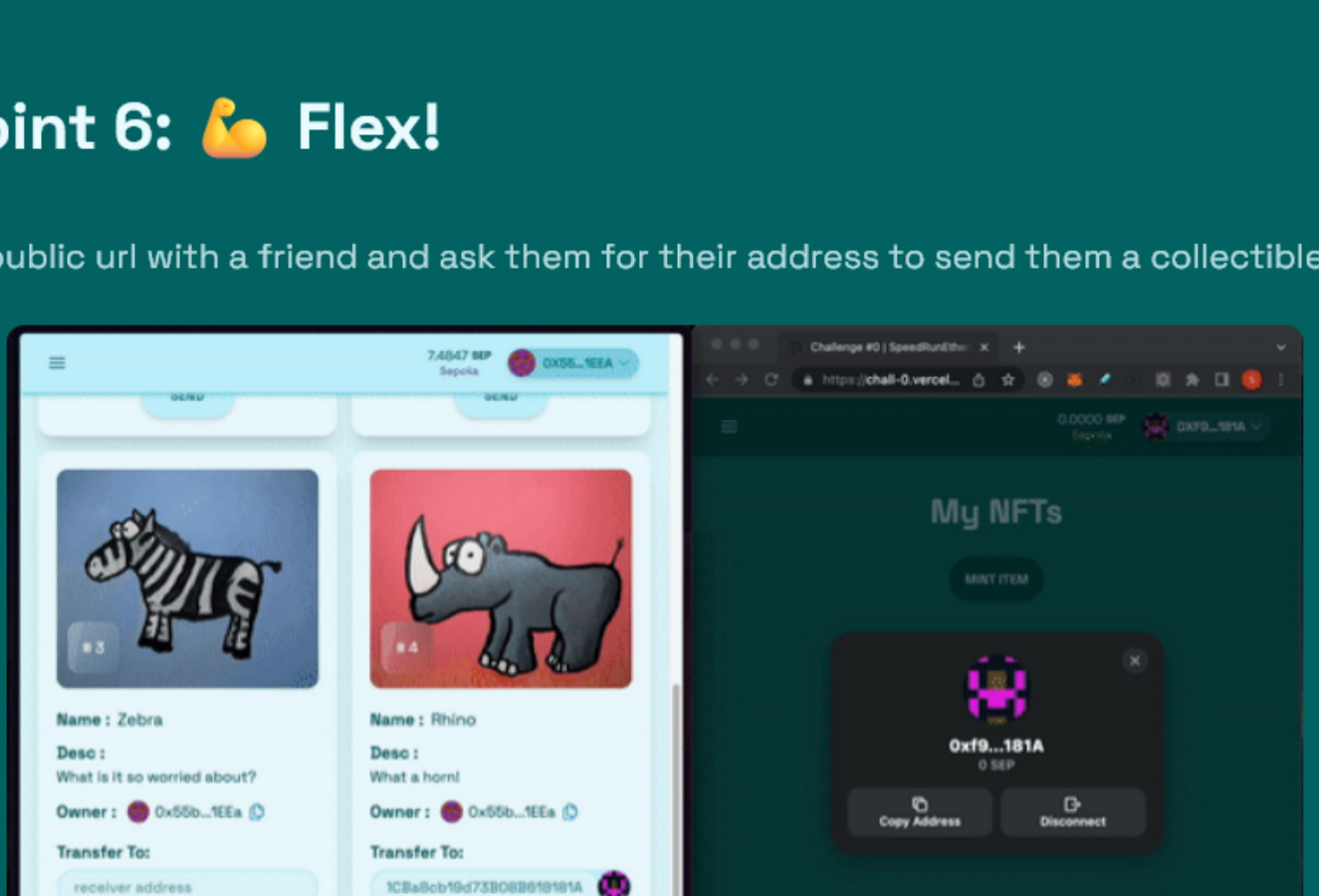
You can verify your smart contract on Etherscan by running (`yarn verify --network network_name`):

```
yarn verify --network sepolia
```

> *"It is okay if it says your contract is already verified. Copy the address of YourCollectable.sol and search it on sepolia Etherscan to find the correct URL you need to submit this challenge."*

Checkpoint 6: 💪 Flex!

👩‍❤️‍👨 Share your public url with a friend and ask them for their address to send them a collectible :)

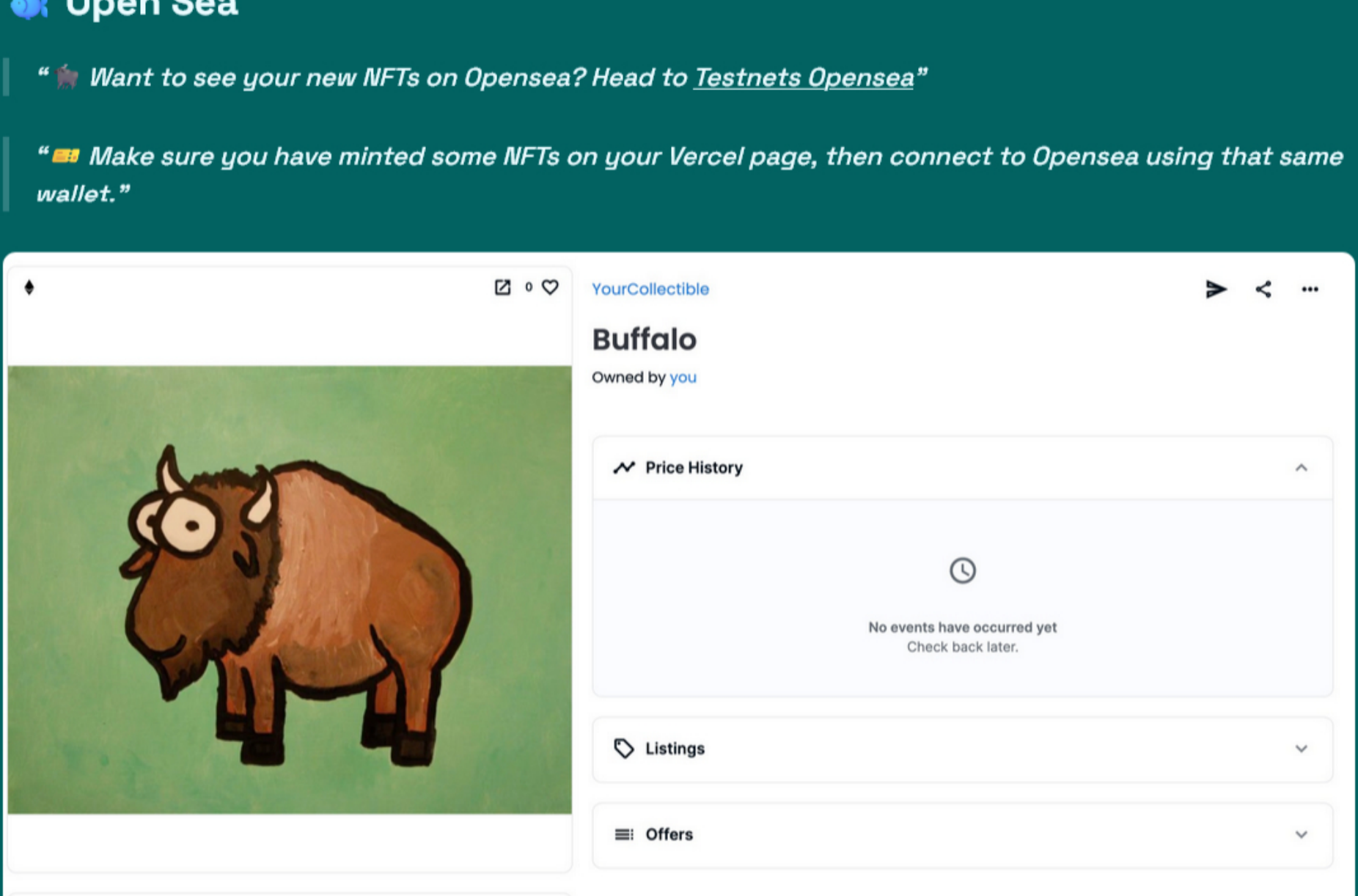

7.4547 SEP
Sepolia
0X55...1EEA
Name : Zebra
Desc :
What is it so worried about?
Owner :
0x55b...1EEa
Transfer To:
receiver address
SEND
Name : Rhino
What a horn!
Challenge #0 | SpeedRunEth
My NFTs
MINT ITEM
0xf9...181A
0 SEP
Copy Address
Disconnect

⚔️ Side Quests

🐟 Open Sea

> *"🐃 Want to see your new NFTs on Opensea? Head to Testnets Opensea"*

> *"🎫 Make sure you have minted some NFTs on your Vercel page, then connect to Opensea using that same wallet."*


YourCollectible
Buffalo
Owned by you
Price History
No events have occurred yet
Check back later.
Listings
Offers
Description

> *"You can see your collection of shiny new NFTs on a testnet!"*

(It can take a while before they show up, but here is an example:) https://testnets.opensea.io/assets/sepolia/0x17ed03686653917efa2194a5252c5f0a4f3dc49c/2

> *"🏃 Head to your next challenge here."*

> *"💬 Problems, questions, comments on the stack? Post them to the 🏗 scaffold-eth developers chat"*

View on GitHub

Related guides

- ERC721 vs. ERC1155: Key Differences, Use Cases & How to Choose
- Mastering ERC721: Developer Guide to NFT Metadata & Best Practices
- NFTs in Web3: Understanding Use Cases Beyond Digital Art
- Solidity NFT Security: 10 Best Practices to Protect Your Collectibles